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NAAC

2727 Counselling Code

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CLASS /SEM : II / III **DATE** : 10.10.2026

BATCH : 05 **PROJECT GUIDE:** Mrs.Deepthi Nair AP/CSE

TEAM MEMBERS :

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ABSTRACT:

AI-BASED EARLY DETECTION OF MACULAR EDEMA:

The AI-Based Retinal OCT Analysis System is designed to assist ophthalmologists and healthcare practitioners in the early, accurate detection of macular edema and retinal pathologies. Optical Coherence Tomography (OCT) is a non-invasive cross-sectional imaging technique essential for diagnosing sight-threatening conditions such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen. During Week 7, the project achieves full system integration, delivering a clinical desktop scanner and an automated diagnostic PDF reporting engine.

The unified platform combines deep transfer learning on DenseNet-121, Explainable AI via Grad-CAM, Monte Carlo Dropout uncertainty estimation, and an Out-of-Distribution input validation gate. Built for practical clinical use, the system provides both real-time webcam scanning for printed records and direct digital file uploading. The diagnostic engine automatically produces standardized, multi-page clinical reports featuring predicted disease probabilities, lesion heatmaps, uncertainty tiers, and ophthalmologist review advisories.

Progress on 10.10.2026 (Week 7)

During this phase, all underlying machine learning modules and computer vision pipelines were consolidated into a unified clinical application. An integrated scanner application was developed to offer dual diagnostic modes: a Webcam Scanning Mode that opens a real-time video feed with an alignment reticle to capture physical OCT printouts, and a File Upload Mode that connects to the operating system's native file dialog to inspect high-resolution digital scans directly. Both modes execute the identical backend pipeline, ensuring rigorous input validation, disease classification, and visual explanation across all inputs.

Additionally, an automated clinical PDF reporting engine was engineered using ReportLab. Upon completing an analysis, the system compiles and exports a structured medical document into a dedicated reports directory. The report incorporates patient scan metadata, categorical diagnostic

probabilities with graphical confidence bars, epistemic uncertainty metrics, side-by-side Grad-CAM lesion heatmaps, and formal clinical disclaimers indicating whether specialist review is recommended. One-click batch startup scripts were created, and comprehensive end-to-end integration tests were completed across all four target classes.

Progress for 24.10.2026

During this upcoming phase, the project will move into the final clinical validation, institutional documentation, and deployment review stage. The system will be evaluated using external clinical datasets obtained through collaboration with Aravind Eye Hospital to benchmark model robustness against real-world imaging variations, noise, and differing OCT machine vendors. Diagnostic sensitivity, specificity, and area under the ROC curve (AUC) will be tabulated across all disease stages.

Furthermore, final academic deliverables will be prepared, including the comprehensive project report, research manuscript, and technical presentation slides. Software packaging, environment dependency freezing, and user operation guides will be finalized to ensure seamless replication and demonstration during final project evaluation.

PROJECT GUIDE PROJECT COORDINATOR HOD